# Yasith Jayawardana, Ph.D.

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## RESEARCH INTERESTS

Deep Learning, Computational Science, HCI, Research Software

## EDUCATION

**Ph.D. in Computer Science, Old Dominion University,** Norfolk VA (4.00 GPA)
(2019 – 2024)

- Dissertation - “StreamingHub, A Realtime Biosignal Processing Framework for Lab Scale Experimentation” (Advisor – Sampath Jayarathna)
- Awards – Dominion Graduate Scholar (2019)

**B.Sc. in Computer Science Engineering, University of Moratuwa,** Sri Lanka (3.84 GPA)
(2015 – 2018)

- Capstone - “MSStack, A Full Stack Microservice Framework with Business Modeling”
- Awards – First Class Honors, Dean’s List (6 of 8 Semesters)

## PROFESSIONAL

**Research Scientist II, Center for AI in Science and Engineering, Georgia Tech,** Atlanta GA
(2025 – Present)

- Building Cybershuttle — a secure platform to connect research compute, code, and data, so researchers can run experiments of any scale and track results from anywhere.

**Researcher (Contract), Center for AI in Science and Engineering, Georgia Tech,** Atlanta GA
(2024 – 2025)

- Built a system to easily switch between local and HPC runtimes on JupyterLab.

**Postgraduate Research Fellow, Center for Advanced Imaging at Harvard,** Cambridge MA
(2022)

- Explored methods to interpret the outcome of single-cell bacteria image classifiers.
- Trained Neural ODEs to approximate the dynamics of multimode fibers.

**Summer Research Intern, Los Alamos National Laboratory,** Los Alamos NM
(2021, 2022)

- Built a platform to extract URLs from scholarly PDFs, create robust links by archiving their content, and replace the original URLs with these robust links.

**Graduate Assistant, Computer Science, Old Dominion University,** Norfolk VA
(2019 – 2024)

- Trained ML methods to diagnose ASD from discrete wavelet coefficients of EEG / IRT data.
- Built a probabilistic estimator to forecast updates to webpages using archived and timestamped content, improving crawl scheduling efficiency.
- TA – Data Structures and Algorithms, Principles of Programming Languages, Computers in Society, Programming with C++, Data Science and Analytics
- Awards - Outstanding Teaching Assistant (2022)

**Developer / Team Lead, DesignBoo (PVT) Ltd**, Sri Lanka
(2018, 2019)

- Developed a React Native app for doctor consultations for The GP Service (UK) Ltd.
- Integrated single sign-on (SSO) and migrated web apps from Angular V2 to V4.
- Set up CI/CD from Bitbucket to AWS and CloudWatch logging, managed DevOps.

**Undergraduate Assistant, Computer Science Engineering, University of Moratuwa,** Sri Lanka
(2018)
- Teaching / Grading – CS2963 Presentation Skills

**Software Engineering Intern, WSO2 (PVT) Ltd,** Sri Lanka
(2017)
- Indexed support docs in Elasticsearch and trained TensorFlow models for query lookup
- Developed an API to answer support questions and integrated it with DialogFlow

**Co-Founder, Onion Inc,** Sri Lanka
(2016 – 2019)
- Developed a Point-of-Sales, Inventory, Reporting, Loan, & Payroll System for Storefronts in Dambulla Economic Centre, Sri Lanka

## VENTURES

**Co-Founder, Marketrix Inc,** San Francisco, CA
(Est 2024)
- Human behavior models that can predict purchase decisions

## PROJECTS

**Cybershuttle** (2024 – Present)
- Building a platform that brings together research data, code, and compute for seamless execution of research workflows.

**Real Time Attention** (2023 – Present)
- Building real-time algorithms to track users' mental effort and attention from eye movements.

**OoD Hypothesis Tests** (2022 – 2024)
- Two-sample hypothesis testing approach leveraging sample homogeneity to enhance out-of-distribution detection for trained models.

**Sparse Imager** (2022)
- Perceiver-style attention model for whole-slide imaging in a sparse multi-resolution format, capturing key regions in high resolution.

**MGaze** (2020 – 2021)
- Web-based platform for multi-user eye tracking via a single webcam.

**Containerized Deployment of Apache OODT** (2019)
- Created Dockerfiles for core components of OODT. Added a Maven Archetype to generate docker/compose files for new OODT projects.

**StreamingHub** (2019 – 2024)
- Biosignal processing framework for assembling computational elements into real-time workflows and processing live and recorded biosignal data streams.

**DengAI** (2017)
- Machine learning model to predict dengue spread using weather and geography features. Awarded 1st Place in DrivenData 2017.

**Raven** (2016 – 2017)
- Platform for Real-time customer satisfaction monitoring for the hospitality industry. Uses CCTV feeds for face tracking, emotion analysis, satisfaction monitoring, and real-time alerts. Awarded 3rd Place in TadHack 2017.

## SERVICES

**Invited Talks**

- Re-Streaming and Synthesis for Real-Time Analysis, WSDL Research Expo, ODU, 2021
- Machine Learning on Data Streams, STRS Camp, ODU, 2020
- Diagnosing ASD from Brain Activity, CS395 Invited Lecture, ODU, 2020
- Data Science with EEG Signals, CS Summer Research Workshop, ODU, 2019

**Peer Review**

- Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- ACM/IEEE Joint Conference on Digital Libraries (JCDL), 2020, 2021, 2022, 2023
- IEEE International Conference on Big Data (BigData), 2023
- ACM Conference on Human Information Interaction and Retrieval (CHIIR), 2020, 2021
- ACM Conference on Intelligent User Interfaces (IUI), 2020, 2021
- ACM Augmented Human International Conference (AH), 2019

**Volunteering**

- Annual Trick or Research Event, ODU, 2019 – 2024
- 34th Great Computer Challenge, ODU, 2019
- Science Connection Day, ODU, 2019
- STEAM on Spectrum, VMASC ODU, 2019

**Leadership**

- Vice President, Sri Lankan Association, ODU, 2022

## TEACHING

- CS250 Problem Solving and Programming II, ODU, 2021
- CS620 Introduction to Data Science and Analytics, ODU, 2021
- CS300T Computers in Society, ODU, 2020
- CS361 Data Structures and Algorithms, ODU, 2020
- CS355 Principles of Programming Languages, ODU, 2020
- CS2963 Presentation Skills, ODU, 2018

## AWARDS

- Outstanding Teaching Assistant, 2022, Old Dominion University
- Sigma Xi Grant in Aid of Research (GIAR), 2020 – Amount: $500
- Dominion Graduate Scholar, 2019, Old Dominion University
- First Class Honors, 2018, University of Moratuwa
- Dean's List (6 of 8 Semesters), 2014-2018, University of Moratuwa

## MEMBERSHIPS

- Graduate Student Member, Honor Society of Phi Kappa Phi (PKP)
- Student Member, ACM
- Student Member, IEEE
- Student Member, IESL

## TECHNICAL EXPERTISE

- **Languages -** Python, Go, Java, C/C++, C#, Julia, PHP, HTML/JS/CSS
- **Frameworks (ML) -** PyTorch, Lightning, TensorFlow, TF.js, OpenCV, Scikit-Learn, SciPy, NumPy, Pandas
- **Frameworks (Web) -** Node.js, Angular, React, React-Native, Flask
- **Databases -** MySQL, PostgreSQL, MongoDB, SQLite, Redis, DynamoDB
- **Cloud -** Docker, Kubernetes, S3, RDS, EC2, ELB, EKS

- **Biosensor Software -** Pupil Capture, EEGLab, Node-RED, LabStreamingLayer, WEKA, NeuroPype, Orange
- **Biosensor Hardware -** PupilLabs Core, Empatica E4, Emotiv Insight, CGX Quick-30

## PUBLICATIONS

### Journal Articles

[2025] G Jayawardena, **Y Jayawardana**, J Gwizdka, "Measuring Mental Effort in Real Time Using Pupillometry," In Journal of Eye Movement Research (JEMR), MDPI.

[2024] Y Abeysinghe, B Mahanama, G Jayawardena, **Y Jayawardana**, M Sunkara, AT Duchowski, VG Ashok, and S Jayarathna, "A-DisETrac Advanced Analytic Dashboard for Distributed Eye Tracking," In International Journal of Multimedia Data Engineering and Management (IJMDEM), IGI Global.

[2022] B Mahanama, **Y Jayawardana**, S Rengarajan, G Jayawardena, L Chukoskie, J Snider, S Jayarathna, "Eye Movement and Pupil Measures: A Review", In Frontiers in Computer Science, Frontiers Media**. [Editor's Pick]**

[2019] S Jayarathna, **Y Jayawardana**, M Jaime, S Thapaliya, "Electroencephalogram (EEG) for Delineating Objective Measure of Autism Spectrum Disorder", in Computational Models for Biomedical Reasoning and Problem Solving, IGI Global.

### Conference Proceedings

[2025] **Y Jayawardana,** S Jayarathna, DN Wadduwage**,** "Thinking in Groups: Permutation Tests Reveal Near-Out of Distribution". **[In Review]**

[2025] G Jayawardena, **Y Jayawardana**, S Jayarathna, "A Real-Time Approach to Capture Ambient and Focal Attention in Visual Search", in ACM Symposium on Eye Tracking Research (ETRA).

[2024] **Y Jayawardana,** D Wannipurage, E Abeysinghe, and S Marru, "Enhancing Research Productivity: Seamless Integration of Personal Devices and HPC Resources with the Cybershuttle Notebook Gateway" in Practice and Experience in Advanced Research Computing: Human Powered Computing (PEARC). ACM.

[2023] H Kariyawasam, S Jayasinghe, **Y Jayawardana**, DN Wadduwage, "Differentiable Modeling of Nonlinear Dynamics in Multimode Fibers" in IEEE Photonics Conference (IPC), IEEE.

[2022] **Y Jayawardana,** VG Ashok, S Jayarathna, "StreamingHub: Interactive Stream Analysis Workflows" in ACM/IEEE Joint Conference on Digital Libraries (JCDL), ACM.

[2022] G Jayawardena, **Y Jayawardana**, S Jayarathna, J Högström, T Papa, D Akkil, AT Duchowski, V Peysakhovich, I Krejtz, N Gehrer, K Krejtz. "Toward a real-time index of pupillary activity as an indicator of cognitive load" in Procedia Computer Science.

[2021] **Y Jayawardana,** G Jayawardena, AT Duchowski, S Jayarathna, "Metadata-Driven Eye Tracking for Real-Time Applications" in ACM Symposium on Document Engineering (DocEng), ACM.

[2020] **Y Jayawardana,** AC Nwala, G Jayawardena, J Wu, S Jayarathna, ML Nelson, CL Giles, "Modeling Updates of Scholarly Webpages Using Archived Data" in IEEE International Conference on Big Data (Big Data), IEEE.

[2020] **Y Jayawardana,** S Jayarathna, "Streaming Analytics and Workflow Automation for DFS" in ACM/IEEE Joint Conference on Digital Libraries (JCDL), ACM.

[2020] B Mahanama, **Y Jayawardana**, S Jayarathna, "Gaze-Net: Appearance-Based Gaze Estimation using Capsule Networks" in Augmented Human Conference (AH), ACM.

[2019] D Haputhanthri, G Brihadiswaran, S Gunathilaka, D Meedeniya, **Y Jayawardana**, S Jayarathna, M Jaime, “An EEG based Channel Optimized Classification Approach for Autism Spectrum Disorder” in Multidisciplinary Engineering Research Conference (MERCon), IEEE.

[2019] **Y Jayawardana,** S Jayarathna, “DFS: A Dataset File System for Data Discovering Users”, in ACM/IEEE Joint Conference on Digital Libraries (JCDL), ACM **[Best Poster Nominee]**

[2019] **Y Jayawardana,** M Jaime, S Jayarathna, “Analysis of Temporal Relationships between ASD and Brain Activity through EEG and Machine Learning”, in International Conference on Information Reuse and Integration (IRI), IEEE. **[Best Student Paper]**

[2018] **Y Jayawardana,** R Fernando, G Jayawardena, D Weerasooriya, I Perera, “A Full Stack Microservices Framework with Business Modelling” in International Conference on Advances in ICT for Emerging Regions (ICTer), IEEE.

**Dissertation**

[2024] **Y Jayawardana** “StreamingHub: A Realtime Biosignal Processing Framework for Lab Scale Experimentation”, Doctor of Philosophy, Computer Science, Old Dominion University.

**Preprints / Symposiums**

[2024] **Y Jayawardana,** A Ahmad, BS Ahluwalia, R Ahmad, S Jayarathna, DN Wadduwage**,** “Hypothesis-Driven Deep Learning for Out of Distribution Detection”. **[arXiv]**

[2022] **Y Jayawardana,** BJ Cain, M Klein, SM Jones, “PDF Server: Robustifying URL References in PDF Documents,” Los Alamos National Laboratory (LANL). **[Symposium]**

[2020] **Y. Jayawardana,** S. Jayarathna, “Towards a Consistent Metadata Format for Biosignal Data Analysis”, in ACM/IEEE Joint Conference on Digital Libraries (JCDL). **[Symposium]**

**Newsletters and Press Coverage**

[2024] Marketrix Revolutionizes In-App Product Support with Groundbreaking AI | Accesswire

[2020] Yasith Jayawardana Receives Sigma-Xi Grant | Old Dominion University